Analysis Approaches for the Biomarker-by-Drug Interaction in Aggregated N-of-1 Trials: A Catalog

2026-10-04 16:41 PDT

Author. pmsimstats team

Purpose. This paper lists every analysis approach for the biomarker-by-drug interaction that has been used, evaluated or proposed in this project. It covers:

- the published analysis of Hendrickson et al. (`orig`, commit `58b32a9`) and its options;
- the nine specifications of manuscript 02;
- the test procedures of manuscripts 08 and 10, and the analysis strategies of manuscript 06;
- the summary-measures statistics of `docs/37`;
- the alternatives discussed in `docs/40`, `docs/41` and `docs/42`;
- the analyses in the covariance study (`08-covariance-study.R`), including its follow-up run;
- the 14 analyses of the stress test (`docs/46`, `11-strawman-stress-test.R`), added on 2026-10-07 (Section 3.9).

The approaches are organized along the dimensions on which an analysis can vary (Section 2). The named approaches are then listed with their source, their status, and what is known about the size of their test (Sections 3 and 4). Section 5 summarizes which approaches are known to hold their size, and Section 6 lists the gaps.

Status note (2026-10-07). The catalog was written on 2026-10-04, while the covariance-study follow-up was running. Its status entries have been updated with dated notes: the follow-up is reported in `docs/45`, and the stress test of `docs/46` has since evaluated 14 analyses in 41 cells. Two statements of the original are corrected where they occur: the unstructured covariance with Kenward-Roger is not correctly sized everywhere, and the flexible-time size excess is confined to model-based and Kenward-Roger inference.

(Updated 2026-10-09: results at $t_{1/2} > 0$ now use anchored carryover; see `docs/53`. Values changed by at most 0.002 (the `docs/40` time-adjustment sizes of O1 and D1); conclusions unchanged. The covariance-study and stress-test entries still report the runs with the published recursive carryover, whose anchored reruns are pending.)

Contents

Notation index and glossary 4

1. Summary 5

2. Dimensions of an analysis 6

    2.1 Estimator family 6

    2.2 Drug-exposure coding 7

    2.3 Time adjustment 8

    2.4 Expectancy 8

    2.5 Baseline handling 9

    2.6 Random effects 10

    2.7 Working covariance of the residuals 10

    2.8 Standard error and degrees of freedom 11

    2.9 Path or sequence term 12

    2.10 Biomarker coding and target 12

    2.11 Design-specific branches 13

3. Named approaches 13

    3.1 The published analysis and its options (orig, 58b32a9) 13

    3.2 Manuscript 02 (carryover specifications; package construct) . 14

    3.3 Manuscripts 08 and 10 (test procedures and calibration; package construct) 15

    3.4 Manuscript 06 (component decomposition) 15

    3.5 Summary measures (`docs/37`) 16

    3.6 This discussion (`docs/40` to `docs/42`; published CS construct) 16

    3.7 The covariance study: main run (published CS and configuration C at $\rho = 0.7$) 17

    3.8 The covariance study: follow-up run (completed; same trials) 18

    3.9 The stress test (`docs/46`; added 2026-10-07) 19

4. What is known about size 21

5. Approaches correctly sized in every cell tested 22

6. Gaps and the evaluation in progress 23

7. Sources 24

List of Tables

1 Notation index and glossary of symbols used in this document . . 4

2 Estimator family options with source and evaluation status . . . 6

3 Drug-exposure coding options E1 to E7 with source and status . 7

4 Time adjustment options with source and evaluation status . . . 8

5 Expectancy modeling options with source and evaluation status . 9

6 Baseline handling options with source and evaluation status . . . 9

7 Random-effects structures with source and evaluation status . . . 10

8 Working residual covariance options with source and evaluation status . . . 11

9 Standard error and degrees-of-freedom options with source and status . . . 11

10 Path or sequence term options with source and evaluation status 12

11 Biomarker coding and target coefficient options with source and status . . . 12

12 Design-specific analysis branch for the OL design 13

13 Published analysis O1 and its options O2 to O5, with evidence on size . . . 13

14 Manuscript 02 carryover specifications G1 to G9 with evidence on size . . . 14

15 Manuscript 08 and 10 test procedures with evidence on size . . . 15

16 Manuscript 06 component-decomposition strategies C1 to C4 with evidence . . . 15

17 Summary-measures statistics S1 to S3 with status and evidence . 16

18 Approaches D1 to D7 from docs/40 to docs/42 with evidence on size . . . 16

19 Covariance study main-run approaches V1 to V7 with size ranges 17

20 Covariance study follow-up approaches F1 to F7, their questions and answers . . . 18

21 Stress-test analyses with null rejection rates and screen outcomes 19

22 Approaches correctly sized in every cell tested, with relative power 23

Notation index and glossary

Notation follows `analysis/report/NOTATION.md`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
$\beta_{bm:D}$	biomarker-by-drug interaction, the estimand	canonical
D_{it}, $D_{bc,it}$	binary drug state (`Db`); exposure-decayed drug indicator (`Dbc`)	canonical
λ, t_{sd}, $t_{1/2}$	decay rate; time since discontinuation; carryover half-life	canonical
$t^a_{1/2}$	half-life assumed by the analysis (`docs/45`)	local
L_{it}	lagged just-off-drug indicator (E3)	local
η	measured expectancy covariate (manuscript 06)	canonical
c_{bm}	biomarker-BR coupling in the stress-test constructs	canonical
B_i	biomarker (code `bm`; centered, `bmc`, in the stress test)	canonical
β_B	biomarker main effect (`docs/41`)	local
ρ	AR(1) serial correlation	canonical
N	total sample size across paths	canonical

Identifiers. The IDs of Section 3 are local to this catalog. The stress test of `docs/46` uses its own IDs (S, A, C1, C2, DS, DA, ES, EA, F1, F2, R, G1, G2, F1A), several of which collide with IDs here (C1, C2, F1, F2, G1, G2). Following the NOTATION convention for paper-local labels, they are cited here with a prefix: 46-F1 is the stress test's F1, and F1 alone is the covariance-study follow-up arm of Section 3.8.

- $\beta_{bm:D}$**.** The biomarker-by-drug interaction, the estimand.
- **`Db`, `Dbc`.** Binary drug state, D_{it}; exposure-decayed drug indicator, $D_{bc,it}$ (1 on drug, $e^{-\lambda t_{sd}}$ off drug).
- **Size.** Type I error of the interaction test; nominal 0.05. An approach is *correctly sized* in a cell if its null rejection rate is within about 2 Monte Carlo standard errors of 0.05.
- **CS, AR(1).** Compound symmetry; first-order autoregressive correlation.
- **CR2.** Bias-reduced cluster-robust (sandwich) standard error, with Satterthwaite (Bell-McCaffrey) degrees of freedom.
- **KR.** Kenward-Roger standard-error and degrees-of-freedom correction.
- **MD.** Mancl-DeRouen bias-corrected sandwich for GEE.
- **cLDA.** Constrained longitudinal data analysis: baseline kept in the response with a common mean.

- **Construct.** The data-generating process a result was obtained under. *Published CS* is the 58b32a9 construct (`docs/36`); *configuration C* is separable AR(1) with graded coupling, at $\rho = 0.7$ in the covariance study and the stress test (the covariance construct of the refined process of paper 01; both studies kept the published recursive carryover, so neither ran the refined process in full); *package* is the current package construct used by manuscripts 02, 06, 08 and
10.

1. Summary

1. **An analysis is a combination of about a dozen choices.** They are: the estimator family, the exposure coding, the time adjustment, the expectancy term, the baseline handling, the random effects, the working covariance, the standard error and degrees of freedom, the path term, the biomarker coding, the target coefficient, and the design-specific branches (Section 2).
2. **About 60 distinct approaches** have been used, evaluated or proposed (Section 3). Most vary one or two choices against a reference.
3. **Correct size depends mainly on the working covariance and the small-sample correction**, not on the exposure coding, the time adjustment or the baseline coding. Each of those moves the Type I error by at most about 0.02. The covariance and its correction move it by up to 0.10 (Section 4).
4. **Known to hold their size in every construct tested:**

   - an unstructured covariance with Kenward-Roger, baseline as a row (covariance study: 0.046 to 0.072);
   - CR2 standard errors with a random intercept and `corCAR1` (papers 02 and 10: 0.045 to 0.049);
   - GEE with the Mancl-DeRouen correction (papers 08 and 10).

   Known to fail somewhere:

   - the published random intercept under AR(1) data (Type I error 0.14 to 0.16 in CO);
   - any unstructured covariance with Satterthwaite only (0.072 to 0.104);
   - naive GEE (1.5 to 1.9 times nominal);
   - flexible time adjustments in Hybrid under carryover (about 0.07, under every covariance tested).

   Corrected (2026-10-07). The unstructured covariance with Kenward-Roger is slightly liberal over the eight covariance-study null cells (mean 0.058; `docs/45`, Section 4.1) and fails the stress test of `docs/46`, where it is 46-A: pooled null rejection 0.057, 0.087 at $N = 35$, and 0.195 to 0.416 in the three original prognostic cells. The flexible-time excess is

real under model-based and Kenward-Roger inference but absent under CR2 (`docs/45`, Section 4.4; `docs/42`, update after Section 1).

5. **Open:** the Kenward-Roger and CR2 versions of the baseline-covariate coding, CR2 with structured covariances under AR(1) data, and the mechanism behind the row coding's power under AR(1). All are in the follow-up run now in progress (Section 6). *Answered (2026-10-07)* by `docs/45`: the covariate coding is slightly liberal with Kenward-Roger (0.059) and nominal with `cs` and CR2 (0.048); CR2 with a parsimonious covariance is nominal in every construct (0.044 to 0.054); the row coding's power under AR(1) comes from the shared biomarker effect β_B (Section 3.8).
6. **The stress test (2026-10-07; `docs/46`, Section 3.9).** 14 analyses, 41 cells, 28,000 trials. Ten survive the size and sensitivity screens: 46-S, 46-C1, 46-C2, 46-DS, 46-ES, 46-EA, 46-F1, 46-F2, 46-G1 and 46-G2. Five of them are also nominal in every prognostic cell: 46-ES, 46-EA, 46-F1, 46-G1 and 46-G2. Among these, 46-G2 and 46-F1 tie on maximin shortfall (0.232 and 0.242), and the rule selects 46-F1, which has fewer covariance parameters: a random intercept with `corCAR1` residuals, CR2 inference, and the expectancy term `De` with `bmc:De` (`docs/49`).

2. Dimensions of an analysis

Each dimension lists its options, where each was used or proposed, and its current status. "Evaluated" means a simulation result exists; the source column gives the paper.

2.1 Estimator family

Table 2: Estimator family options with source and evaluation status

Option	Source	Status
Linear mixed model, conditional (`lmer`, `nlme::lme`)	orig, 02, 06, 08, 10, covariance study	evaluated
Marginal model with structured covariance (MMRM, `mmrm`, `gls`)	covariance study; `docs/41`, `docs/42`	evaluated
Summary measures: per-participant on-minus-off contrast, regressed on the biomarker (E9)	`docs/37`; 02 (G5); 08 (continuous-biomarker contrast)	evaluated
Strict repeated-measures ANOVA, dichotomized biomarker	`docs/26`; 08	evaluated (08)
GEE with sandwich variance	08; 10	evaluated

Option	Source	Status
Two-stage approaches other than E9	08 (literature review)	not evaluated
Nonlinear component-matched model (Gompertz BR, PB, TV)	`docs/31`; 06	partly evaluated (06, recovery study)
Bayesian hierarchical N-of-1 model	02 (named as out of scope)	not evaluated
Randomization or permutation test	08 (literature)	not evaluated
Latent-class mixture model	manuscript 03	evaluated (different estimand)

2.2 Drug-exposure coding

Table 3: Drug-exposure coding options E1 to E7 with source and status

Option	Code	Source	Status
Binary on-drug indicator `Db`	E1	orig; 02 G1	evaluated
Exposure-decayed `Dbc`, assumed half-life	E2	02 G3; implementations/original	evaluated
Binary plus lagged just-off-drug term L_{it}	E3	02 G2 (Jones-Kenward)	evaluated: no gain over E1
Binary plus linear time-since-discontinuation	E4	implementations/original (`simplecarryover`)	retired in 02
Lag term crossed with biomarker (L_{it} and $B_i L_{it}$)	E5	02, earlier S7	retired: worse than E1
t_{sd} crossed with biomarker	E6	02, earlier S7	retired: worse than E1
`Dbc` with half-life chosen by AIC	E7	02 G4; stress test (46-DS, 46-DA, 46-F1A)	evaluated: nominal in one cell (02); in the stress test 46-DS passes, 46-DA and 46-F1A fail size (pooled 0.065 and 0.057)
`Dbc` with alternative decay forms (Weibull, linear, power)		02 sensitivity blocks; NOTATION	evaluated as misspecification

2.3 Time adjustment

Table 4: Time adjustment options with source and evaluation status

Option	Source	Status
None	E9 (`docs/37`)	evaluated
Linear `t`	orig; 02; all mixed models above	evaluated
Linear `t` plus expectancy `De`	orig option (`useDE`); `docs/40`; stress test (46-F1, with `bmc:De`)	evaluated (2026-10-07): 46-F1 nominal and prognostic-robust; the selected analysis
Natural spline in `t`, 3 df	`docs/40`, `docs/42`; covariance study	evaluated: size excess in the Hybrid carryover cell under model-based and Kenward-Roger inference, none under CR2
Visit as factor	`docs/40`, `docs/42`; covariance study	evaluated: as the spline
Biomarker by linear time (`bmc:t`)	stress test (46-F2)	evaluated (2026-10-07): nominal, but fails the expectancy-prognostic cell
Phase indicators, phase by drug (attribution)	06 (phase-augmented)	evaluated (06): no bias reduction, less power
Random slope on `t`	orig option (`t_random_slope`)	not evaluated
Path-specific intercepts (E9 stage 2)	`docs/37` (stratified E9)	closed form only

2.4 Expectancy

Table 5: Expectancy modeling options with source and evaluation status

Option	Source	Status
Not modeled	orig as published	evaluated
Design expectancy `De` as a fixed effect	orig option	population-mean limit (`docs/40`); evaluated with `bmc:De` in the stress test (46-F1, 46-G1, 46-G2, 46-F1A; 2026-10-07)
Absorbed by visit effects	`docs/40`, `docs/42`	evaluated
Measured belief covariate η	06 (balanced-placebo design)	evaluated (06)

2.5 Baseline handling

Table 6: Baseline handling options with source and evaluation status

Option	Source	Status
Ninth response row, coded off drug	orig	evaluated
Excluded, no covariate	02 (all nine)	evaluated (02)
Covariate (ANCOVA)	`docs/41`; covariance study	evaluated with `cs` and `us`: Satterthwaite (main run), Kenward-Roger and CR2 (follow-up, `docs/45`)
cLDA: row with common baseline mean, unstructured covariance	`docs/41`	equivalent to the covariate coding only with β_B freed at baseline (`docs/45`, Section 6)
Change from baseline	`docs/41`	identical to ANCOVA in the construct

Option	Source	Status
Baseline-by-drug moderator	`docs/41`	not evaluated (zero in the construct)
Row, with a separate biomarker effect at baseline	follow-up run (`us-KR-sepbm`); stress test (46-ES, 46-EA)	evaluated (2026-10-07): 46-ES and 46-EA nominal and prognostic-robust

2.6 Random effects

Table 7: Random-effects structures with source and evaluation status

Option	Source	Status
None (marginal covariance only)	MMRM; 10 (M0, M2)	evaluated
Random intercept	orig; 02; 10 (M1, M3)	evaluated
Random intercept and slope on `t`	orig option	not evaluated
Random intercept, post-baseline shift and expectancy coefficient, phase-specific residual variance	this discussion (implied by the published construct)	partly evaluated (2026-10-07): 46-G1 and 46-G2 have a fixed expectancy term and phase-specific residual variances, without the random shift or random expectancy coefficient

2.7 Working covariance of the residuals

Table 8: Working residual covariance options with source and evaluation status

Option	Source	Status
Independent	orig; 10 (M1)	evaluated
`corCAR1` (continuous-time AR(1))	02; 06; 08; 10 (M2, M3)	evaluated
Compound symmetry `cs`	covariance study (equals orig's random intercept)	evaluated
Heterogeneous CS `csh`	covariance study	evaluated
Unstructured `us`	covariance study; 10	evaluated
Spatial exponential `sp_exp`	covariance study	evaluated
AR(1), Toeplitz, ante-dependence	`mmrm` options	not evaluated (assume equal spacing)
Phase-specific variance (`varIdent`)	this discussion; stress test	evaluated (2026-10-07): 46-C1, 46-C2, 46-G1, 46-G2
GEE working correlation: independence, exchangeable, AR(1)	08; 10 (AR(1))	evaluated

2.8 Standard error and degrees of freedom

Table 9: Standard error and degrees-of-freedom options with source and status

Option	Source	Status
Model-based SE, Satterthwaite df (`lmerTest`)	orig	evaluated
Model-based SE, containment df (`nlme`)	02; 06; 08	evaluated
Model-based SE, Satterthwaite df (`mmrm`)	covariance study	evaluated
Kenward-Roger	covariance study (`us`); 10	evaluated for `us`, `cs`, `csh` and the baseline covariate (`docs/45`): repairs `us` only

Option	Source	Status
CR2 sandwich, Satterthwaite df (`clubSandwich`)	02 (G6 to G9); 10; stress test	evaluated; the inference of 46-S, 46-C1, 46-DS, 46-ES, 46-F1, 46-F2, 46-G1 and 46-F1A
CR2 sandwich, Satterthwaite df (`mmrm`)	follow-up run	evaluated (`docs/45`): nominal with parsimonious covariances, liberal with `us` (0.086)
CR0, CR1, CR3 sandwiches	10 (discussed)	not evaluated
Naive GEE sandwich	08; 10	evaluated
Mancl-DeRouen GEE sandwich	08; 10	evaluated
OLS t, $N - 2$ df	E9	evaluated

2.9 Path or sequence term

Table 10: Path or sequence term options with source and evaluation status

Option	Source	Status
None	orig and most	evaluated
Fixed path effect	`docs/37`, single-path study	evaluated: no effect on `lmer` power
Path-specific intercepts in E9	`docs/37`	closed form only

2.10 Biomarker coding and target

Table 11: Biomarker coding and target coefficient options with source and status

Option	Source	Status
Uncentered biomarker	orig; 02	evaluated
Centered biomarker	`docs/40`; covariance study	evaluated (target unchanged)

Option	Source	Status
Dichotomized biomarker	08 (strict RM-ANOVA)	evaluated: costs 0.08 to 0.21 power
Target `bm:Db`	orig; 02 G1, G2	evaluated
Target `bm:Dbc`	02 G3, G4	evaluated
Component-specific slopes	06	evaluated
Joint 2-df test (E5, E6)	02	retired

2.11 Design-specific branches

Table 12: Design-specific analysis branch for the OL design

Option	Source	Status
Biomarker-by-time test when no participant varies in drug state (OL design): `Sx ~ bm + t + bm*t`	orig	evaluated in the published results; confounded with natural course and expectancy

3. Named approaches

Each row is a complete specification that has been fitted or proposed. Status is *evaluated*, *running* or *proposed*; the evidence column gives the construct and the key size result where one exists.

3.1 The published analysis and its options (orig, 58b32a9)

Table 13: Published analysis O1 and its options O2 to O5, with evidence on size

ID	Specification	Status	Evidence on size
O1	`lmer`, fixed part `Sx ~ bm + Db + t + bm*Db`, random intercept by `ptID`, baseline row, Satterthwaite (published default)	evaluated	published CS: 0.049 to 0.056 (Hybrid), 0.04 to 0.05 (CO). Published results: 0.01 to 0.09. AR(1) data: 0.14 to 0.23 in CO (`docs/36`)

ID	Specification	Status	Evidence on size
O2	O1 plus `De` (`useDE = TRUE`)	proposed	dropped by Hendrickson et al. for collinearity; correlation with `Db` 0.59 (`docs/40`). Not simulated as specified; a CR2 analogue with `bmc:De` is 46-F1
O3	O1 with a random intercept and random slope on `t` by `ptID` (`t_random_slope = TRUE`)	proposed	
O4	OL branch: fixed part `Sx ~ bm + t + bm*t`, random intercept by `ptID`	evaluated	confounded test; not a drug-by-biomarker interaction
O5	Later vendored implementation: `nlme::lme(Sx ~ bm + t + Dbc + bm:Dbc)`, random intercept, `corCAR1`	evaluated as 02 G3	see 02 G3

3.2 Manuscript 02 (carryover specifications; package construct)

All use `nlme::lme` with a random intercept and `corCAR1`, no baseline row, linear `t`.

Table 14: Manuscript 02 carryover specifications G1 to G9 with evidence on size

ID	Specification	Status	Evidence on size
G1	`Db`, target `bm:Db`	evaluated	mildly conservative (mean null 0.036 to 0.039)
G2	G1 plus lag L_{it}	evaluated	as G1; no power gain over G1
G3	`Dbc`, target `bm:Dbc`	evaluated	as G1
G4	G3, half-life chosen by AIC	evaluated (one cell)	nominal despite selection
G5	Paired difference (E9)	evaluated	
G6 to G9	G1, G2, G3, G4 with CR2	evaluated	CR2 restores nominal size (0.045 to 0.049 in 10)

3.3 Manuscripts 08 and 10 (test procedures and calibration; package construct)

Table 15: Manuscript 08 and 10 test procedures with evidence on size

ID	Specification	Status	Evidence on size
T1	Strict RM-ANOVA, dichotomized biomarker	evaluated (08)	dichotomization costs 0.08 to 0.21 power
T2	Continuous-biomarker within-participant contrast	evaluated (08)	calibrated
T3	`lme`, random intercept, `corCAR1` (production)	evaluated (08, 10 M3)	conservative, 0.029 to 0.033 (10)
T4	GEE, naive sandwich	evaluated (08, 10)	anticonservative, 1.5 to 1.9 times nominal (08); 0.067 to 0.069 (10)
T5	GEE, Mancl-DeRouen	evaluated (08, 10)	restores nominal size
M0	OLS, no within-participant correlation	evaluated (10)	not anticonservative
M1	Random intercept only	evaluated (10)	well calibrated in 10's construct
M2	`corCAR1` only	evaluated (10)	intermediate
M3	Random intercept plus `corCAR1`	evaluated (10)	conservative; CR2 fixes it; KR inert on it
M-us	Unstructured, KR against Satterthwaite	evaluated (10)	Satterthwaite 0.076, KR 0.059

3.4 Manuscript 06 (component decomposition)

Table 16: Manuscript 06 component-decomposition strategies C1 to C4 with evidence

ID	Specification	Status	Evidence
C1	One-component (`bm:Dbc`)	evaluated	unbiased if the biomarker is unrelated to PB and TV; size 0.031 to 0.056
C2	Phase-augmented: phase, phase by drug, phase by biomarker by drug	evaluated	no bias reduction; less power; size 0.038 to 0.049

ID	Specification	Status	Evidence
C3	Belief-covariate decomposition	evaluated	recovers the pharmacological slope only on a balanced-placebo design
C4	Blinded-stratum contrast; Gompertz-basis decomposition	evaluated	fail on the Hybrid design

3.5 Summary measures (`docs/37`)

Table 17: Summary-measures statistics S1 to S3 with status and evidence

ID	Specification	Status	Evidence
S1	E9: post-baseline on-minus-off contrast, OLS on biomarker	evaluated (closed form and Monte Carlo)	exact closed form; validated over 566 cell-variants
S2	E9b: baseline counted as off drug	evaluated	mirrors O1's baseline coding; 15% to 16% noisier than E9 without carryover (`docs/41`)
S3	Stratified E9: path-specific intercepts	closed form only	removes the CO period-effect loss

3.6 This discussion (`docs/40` to `docs/42`; published CS construct)

Table 18: Approaches D1 to D7 from docs/40 to docs/42 with evidence on size

ID	Specification	Status	Evidence on size
D1	O1 with `ns(t, 3)`	evaluated	Hybrid, $t_{1/2} = 1$: 0.075 (O1 0.056); recurs in `docs/45` (0.070), absent under CR2
D2	O1 with visit factor	evaluated	Hybrid, $t_{1/2} = 1$: 0.080; recurs in `docs/45` (0.072), absent under CR2
D3	O1 plus fixed path effect	evaluated	power changes by at most 0.001
D4	O1 on a single path against the mixture	evaluated	no material difference in Hybrid

ID	Specification	Status	Evidence on size
D5	Randomized (multinomial or permuted-block) path allocation	proposed	
D6	Random intercept, post-baseline shift and expectancy coefficient, `varIdent` by phase	proposed; partly evaluated as 46-G1, 46-G2 (fixed, not random, expectancy coefficient)	46-G1 pooled 0.049, 46-G2 0.047
D7	Baseline-by-drug moderator	proposed	

3.7 The covariance study: main run (published CS and configuration C at $\rho = 0.7$)

All `mmrm`, centered biomarker, binary `Db`. Fitted to the same trials: 16 cells, 250 alternative and 500 null replicates.

Table 19: Covariance study main-run approaches V1 to V7 with size ranges

ID	Coding	Covariance	Time	Test	Size range
V1	row	`cs`	linear, spline, visit	Satterthwaite	CS 0.036 to 0.072; configuration C, CO 0.140 to 0.158
V2	row	`csh`	linear, spline, visit	Satterthwaite	as V1 (CO, configuration C: 0.138 to 0.164)
V3	row	`us`	linear, spline, visit	Satterthwaite	0.072 to 0.104
V4	row	`sp_exp`	linear, spline, visit	Satterthwaite	0.026 to 0.062; low power under CS

ID	Coding	Covariance	Time	Test	Size range
V5	row	`us`	linear, visit	Kenward-Roger	0.046 to 0.072
V6	covariate	`cs`	linear, spline, visit	Satterthwaite	CS 0.034 to 0.064; configuration C, CO 0.146 to 0.164
V7	covariate	`us`	linear, spline, visit	Satterthwaite	0.072 to 0.096

3.8 The covariance study: follow-up run (completed; same trials)

The run was in progress when this catalog was written. The last column was added on 2026-10-07 from `docs/45` (Section 4.1 mean sizes over the eight null cells, linear time; Section 6 for F7).

Table 20: Covariance study follow-up approaches F1 to F7, their questions and answers

ID	Coding	Covariance	Time	Test	Question	Answer (`docs/45`)
F1	covariate	`us`	linear, spline, visit	Kenward-Roger	is the covariate coding correctly sized?	slightly liberal, 0.059
F2	covariate	`cs`	linear, visit	Kenward-Roger	does KR change a parsimonious model?	no; liberal, 0.077
F3	row	`us`	spline	Kenward-Roger	completes V5	same class as linear time, within 0.005
F4	row	`cs`, `csh`	linear, visit	Kenward-Roger	can KR repair a misspecified covariance?	no; 0.079 and 0.073

ID	Coding	Covariance	Time	Test	Question	Answer (`docs/45`)
F5	row	`cs`, `csh`, `us`, `sp_exp`	linear, visit	CR2	does CR2 hold size under every covariance?	nominal for `cs`, `csh`, `sp_exp` (0.048 to 0.054); liberal for `us` (0.086)
F6	covariate	`cs`, `us`	linear, visit	CR2	as F5, covariate coding	`cs` nominal (0.048); `us` liberal (0.080)
F7	row, separate baseline biomarker effect	`us`	visit	Kenward-Roger	does the shared biomarker effect explain the row coding's power?	yes: freed, power matches the covariate coding (0.780 against 0.792)

3.9 The stress test (`docs/46`; added 2026-10-07)

All 14 analyses were fitted to the same simulated Hybrid trials: 41 cells (39 in the original working protocol, plus two prognostic-baseline cells, 40 and 41), 15 null cells of 1,000 replicates and 26 alternative cells of 500, 28,000 trials in all. The base cell is configuration C at $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week, with graded coupling, the positive-definite stop and the published recursive carryover (inspected in `11-strawman-stress-test.R`). Every analysis has linear time and a centered biomarker (`bmc`), and uses $D_{bc,it}$ at $t^a_{1/2} = 0.5$ week, except 46-R (binary D_{it}) and the AIC arms 46-DS, 46-DA and 46-F1A (half-life chosen over $\{0, 0.25, 0.5, 1, 2\}$). IDs carry the 46- prefix of the Identifiers note above.

Table 21: Stress-test analyses with null rejection rates and screen outcomes

ID	Specification	Pooled size	Worst non-prognostic cell	Prognostic cells (BL, PB, TV constant, TV growing)	Outcome
46-S	random intercept, `corCAR1`, CR2, baseline row	0.048	0.059	0.050, 0.100, 0.099, 0.049	survives; flagged

ID	Specification	Pooled size	Worst non-prognostic cell	Prognostic cells (BL, PB, TV constant, TV growing)	Outcome
46-A	`us`, Kenward-Roger, baseline row (`docs/45` primary)	0.057	0.087 ($N = 35$)	0.051, 0.222, 0.416, 0.195	fails size
46-C1	46-S with `varIdent` by phase, CR2	0.053	0.069	0.054, 0.205, 0.382, 0.173	survives; flagged
46-C2	46-C1, model-based	0.052	0.066	0.050, 0.193, 0.393, 0.172	survives; flagged
46-DS	46-S, $t^a_{1/2}$ chosen by AIC	0.053	0.068	0.055, 0.112, 0.115, 0.058	survives; flagged
46-DA	46-A, $t^a_{1/2}$ chosen by AIC	0.065	0.103 ($N = 35$)	0.064, 0.256, 0.509, 0.245	fails size
46-ES	46-S with β_B freed at baseline	0.048	0.059	0.050, 0.049, 0.049, 0.063	survives; robust
46-EA	46-A with β_B freed at baseline	0.052	0.070	0.066, 0.050, 0.049, 0.057	survives; robust
46-F1	46-S plus `De + bmc:De`	0.051	0.062	0.040, 0.040, 0.047, 0.046	survives; robust; **selected**
46-F2	46-S plus `bmc:t`	0.049	0.059	0.048, 0.098, 0.090, 0.048	survives; flagged
46-R	published reference: compound symmetry (`mmrm cs`, equivalent to a random intercept), model-based SE with Satterthwaite df, binary D_{it}	0.069	0.081	0.075, 0.159, 0.107, 0.053	fails size
46-G1	46-F1 mean model, phase-specific residual variances, CR2	0.049	0.060	0.042, 0.042, 0.055, 0.044	survives; robust

ID	Specification	Pooled size	Worst non-prognostic cell	Prognostic cells (BL, PB, TV constant, TV growing)	Outcome
46-G2	46-G1, model-based	0.047	0.066	0.046, 0.045, 0.062, 0.043	survives; robust
46-F1A	46-F1, $t^a_{1/2}$ chosen by AIC	0.057	0.073 ($t_{1/2} = 1$)	0.045, 0.047, 0.059, 0.052	fails size

Pooled size is over the 11 non-prognostic null cells (11,000 trials); the nominal band is 0.0458 to 0.0542 (2 MCSE) with no cell above 0.075. *Flagged* means the analysis fails at least one prognostic cell; *robust* means it is nominal in all four. All 14 pass the monotonicity screen; the symmetry screen was dropped in Amendment 1 of `docs/46`, because power is symmetric in the sign of c_{bm} by construction. Maximin shortfall of size-adjusted power over the non-prognostic alternative cells: 46-C1 0.034, 46-C2 0.040, 46-S 0.144, 46-F2 0.160, 46-DS 0.182, 46-G2 0.232, 46-F1 0.242, 46-G1 0.252, 46-ES 0.278, 46-EA 0.318. 46-C1 has the smallest maximin but fails the prognostic cells. Among the robust survivors 46-G2 and 46-F1 tie within 0.02; neither reaches power 0.80 anywhere on the power-curve grid, so the tie goes to the analysis with fewer covariance parameters, 46-F1. (Clarified 2026-10-08: "maximin shortfall" is the term used by the decision outputs; selecting the smallest largest shortfall is a minimax-regret rule, and the best analysis in each cell is taken over all ten survivors, the flagged 46-C1 and 46-C2 included, so the robust analyses' shortfalls are measured against the phase-variance models. The 46-R row was also reworded on 2026-10-08: the driver fits it as the `mmrm` compound-symmetry structure with Satterthwaite degrees of freedom, not as an `lme` random intercept with containment degrees of freedom. Neither change alters any rate or the selection. See `analysis/report/compendium-mathematics.tex`, Sections "Size, power screens and the selection criteria of the stress test" and "The stress-test construct and its fourteen analyses".) Failed fits were 0 to 12 per analysis out of 28,000 (at most 0.6% in a cell; verified from the `fails` column of `summary-full*.csv`). The working protocol counted failed fits as non-rejections; the summaries exclude them, which changes no rate by more than 0.0025. (Verified against `analysis/data/quick-sim/carryover-closed-form/strawman-stress/decision-*-ds-nosym-all14.csv`; decision rule rerun with all 14 analyses, 2026-10-07.)

4. What is known about size

Main conclusions:

1. **The covariance and its small-sample correction dominate.** Changing the exposure coding (02), the time adjustment (`docs/40`) or the base-

line coding (covariance study) moves the Type I error by at most about 0.02. Changing the working covariance or the correction moves it by up to 0.10.

2. **A parsimonious covariance that is wrong fails in either direction.** The published random intercept is anticonservative under AR(1) data in CO (0.14 to 0.16). The random intercept with `corCAR1` is conservative in the package construct (0.029 to 0.039). CR2 restores the latter. Kenward-Roger does not, being inert on parsimonious structures (10).
3. **An unstructured covariance needs Kenward-Roger.** With Satterthwaite alone it is anticonservative in every construct tested (0.072 to 0.104 in the covariance study, 0.076 in 10). Kenward-Roger brings it to nominal. *Corrected (2026-10-07):* Kenward-Roger brings it close to nominal but not to it: slightly liberal over the covariance study's null cells (mean 0.058, `docs/45`), and failing the stress test's size criterion (46-A: pooled 0.057, 0.087 at $N = 35$).
4. **Sandwich estimators need small-sample corrections.** Naive GEE is anticonservative. Mancl-DeRouen and CR2 restore the size.
5. **A residual size problem remains for flexible time adjustments in Hybrid under carryover** (about 0.07). It persists under every covariance tested, including `us` with Kenward-Roger, and its cause is unexplained. *Corrected (2026-10-07):* the follow-up run tested CR2, under which the excess is absent (0.058 to 0.060 in that cell); under model-based and Kenward-Roger inference it is real, recurring in a second run at 0.068 to 0.072 (`docs/45`, Section 4.4, whose description of this evidence as weak is itself corrected in `docs/42`). Its cause remains unexplained.
6. **The stress test separates the prognostic-robust analyses (added 2026-10-07).** Every analysis that neither frees β_B at baseline nor adds the expectancy term fails at least one cell in which the biomarker is prognostic for TV or PB (null rejection up to 0.509). Freeing β_B at baseline (46-ES, 46-EA) or adding `De + bmc:De` (46-F1, 46-G1, 46-G2) gives nominal size in every prognostic cell; `bmc:t` (46-F2) repairs only the cell with a growing natural-course association (Section 3.9).

Construct dependence. Results come from three constructs: published CS, configuration C, and the package construct. Approaches have not all been tested in all three. The parsimonious covariances are the most construct-dependent: the random intercept is well calibrated under CS data (10 M1; covariance study V1 under CS) and fails under AR(1) data.

5. Approaches correctly sized in every cell tested

Table 22: Approaches correctly sized in every cell tested, with relative power

Approach	Constructs	Size range	Power relative to the published analysis
Row, `us`, Kenward-Roger, linear or visit (V5)	published CS, configuration C	0.046 to 0.072	similar under CS; +0.10 to +0.12 in Hybrid under configuration C
Random intercept plus `corCAR1`, CR2 (02 G6 to G9; 10)	package	0.045 to 0.049	slightly higher than model-based (recovers its conservatism)
GEE, Mancl-DeRouen (T5)	package	nominal	modest power cost against the mixed model (08)
`sp_exp`, Satterthwaite (V4)	published CS, configuration C	0.026 to 0.062	much lower under CS (0.31 to 0.52 against about 0.70)

V5's upper value of 0.072 occurs with the visit-factor time adjustment in Hybrid at $t_{1/2} = 1$ (published CS), the residual problem of Section 4, point 5. With linear time V5 stays within 0.052 to 0.066.

Corrected (2026-10-07). Two rows of this table do not meet the catalog's own criterion. V5 is slightly liberal (mean 0.058, `docs/45`) and its stress-test counterpart 46-A, the same model with $D_{bc,it}$ at 0.5 week, fails the size criterion (pooled 0.057; 0.087 at $N = 35$) and all three original prognostic cells (0.195 to 0.416). It is therefore not recommended, and `docs/45`'s recommendation of it as the primary analysis is superseded by `docs/46` and `docs/49`. V4 (`sp_exp`, Satterthwaite) is conservative (mean 0.042, minimum 0.028; `docs/45`). The analyses now known to be correctly sized and robust in every stress-test cell, prognostic cells included, are 46-ES, 46-EA, 46-F1, 46-G1 and 46-G2 (Section 3.9), all in the configuration C and compound-symmetry constructs of `docs/46`; 46-F1 is the selected analysis.

6. Gaps and the evaluation in progress

Being answered by the follow-up run (F1 to F7): Kenward-Roger and CR2 for the baseline-covariate coding; whether CR2 rescues the misspecified parsimonious covariances under AR(1) data; whether Kenward-Roger does (expected not); and the mechanism behind the row coding's power under AR(1). *Answered (2026-10-07)* in `docs/45`; see the last column of Section 3.8.

Not yet scheduled (status of each item added 2026-10-07):

1. The cause of the flexible-time size excess in Hybrid under carryover (Section 4, point 5). *Open;* CR2 is known to remove it.
2. The random-effects structure implied by the published construct, with phase-specific variance (D6). *Partly done:* 46-G1 and 46-G2 (fixed expectancy term, phase-specific residual variances).
3. A random slope on time (O3) and the expectancy term (O2) by simulation, not only in the population-mean limit. *Partly done:* the expectancy term with `bmc:De` under CR2 is 46-F1; O3 is open.
4. GEE with Mancl-DeRouen and the summary-measures statistic in the published CS and configuration C constructs, for a direct comparison with V5. *Open.*
5. Exposure-weighted coding (`Dbc`) crossed with the covariance choices. Manuscript 02 evaluated it only under `corCAR1`. *Done:* `docs/45` exposure pass (`us-KR`, `us-CR2`, `RI+CAR1-CR2`; no change in size) and every stress-test analysis except 46-R.
6. Randomized path allocation (D5). *Open.*
7. Power at matched size, for every approach found correctly sized. *Done* for the analyses of `docs/45` and `docs/46`.
8. Bayesian hierarchical models and randomization tests, named in the literature reviews of 02 and 08 and not evaluated. *Open.*

Suggested core comparison. Once the follow-up run is in, a single paired study in the three constructs could cover:

- the approaches correctly sized so far (V5, CR2 variants, GEE with Mancl-DeRouen);
- the published analysis as reference (O1);
- crossed with the exposure coding (`Db` against `Dbc`) and the baseline coding.

That would give one table of size and power from which a primary analysis for a Hybrid trial could be chosen. *Status (2026-10-07):* the exploratory stress test of `docs/46` is that study for the Hybrid design, in configuration C and compound symmetry but not the package construct, and without GEE or the summary-measures statistic. It selected 46-F1 (Section 3.9; `docs/49`).

7. Sources

- `orig`: `github.com/rchendrickson/pmsimstats`, commit `58b32a9`, `R/lme_analysis.R` and vignette 1 (inspected).
- Manuscripts: `analysis/report/02-carryover-sensitivity/`, `06-component-decomposition/`, `08-test-procedure-design-sensitivity/`, `10-interaction-test-calibration/` (read for their specifications and headline results; the numbers quoted are those reported in the manuscripts).

- Project documents: `docs/26`, `docs/31`, `docs/36`, `docs/37`, `docs/40`, `docs/41`, `docs/42`; added 2026-10-07: `docs/45`, `docs/46`, `docs/48`, `docs/49`.
- Covariance study: `analysis/scripts/quick-sim/carryover-closed-form/08-covariance-study.R`; results in `analysis/data/quick-sim/carryover-closed-form/covariance-study/summary-alt250-n` (verified); follow-up completed and reported in `docs/45`.
- Stress test: `analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R` and `12-strawman-stress-decision.R`; results in `analysis/data/quick-sim/carryover-closed-form/` (`decision-*-ds-nosym-all14.csv`, `summary-full*.csv`; verified).